Research and Adaptation of Components from the Persistent OS Phantom for Genode

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Data Persistence

Data persistence: period of time during which data exists and is used

R. Morrison, M. P. Atkinson — "Persistent Languages and Architectures", 1990

Today's dominant model — **two-level store**:

- Data is either *active* (RAM) or *stored* (disk)
- Programmer manually moves data between levels
- Consequences: data loss, I/O boilerplate (≈30% of code)

Orthogonal Persistence

Alternative: the **environment** manages persistence

Three principles:

1. **Independence** — data persistence does not depend on how it is manipulated
2. **Data type orthogonality** — all types can be persisted
3. **Identification** — type does not affect persistence identity

Benefits: up to 30% reduction in code size, more efficient use of disk bandwidth and improved reliability failt tolerance.

PhantomOS

PhantomOS is a non-Unix like OS implementing orthogonal persistence (D. Zavalishin).

- Persistent Virtual Machine (PVM): bytecode VM + POSIX layer
- Snapshots saved periodically via **Copy-on-Write**
- Only changed pages written to disk (incremental)
- Programs paused only briefly during snapshot preparation

Currently ported to the **Genode** OS framework:

- Proven microkernels (NOVA, seL4, Fiasco.OC)
- Capability-based security, minimal TCB

Current snapshot mechanism: **Snapper**

Snapper

File-system-based generational snapshot storage.

Previous approach ("superblock"): snapshots stored as monolithic objects.

+ Fast
− No fault tolerance, no storage recovery

Snapper 2.0:

- Unchanged pages **linked**, not copied across generations
- ext4 + fsck for crash recovery
- Integrity checks (hashes per file)
- Configurable redundancy policy

Problem Statement

PhantomOS snapshots capture the **entire machine state** — including sensitive data present in memory.

Snapshots are stored in **plaintext**. An attacker with physical access can mount and read them directly, or fully reproduce the machine state.

- No prior research on this for PhantomOS
- No ready-made solution exists in Genode

Goal: implement a mechanism for the secure storage of snapshots.

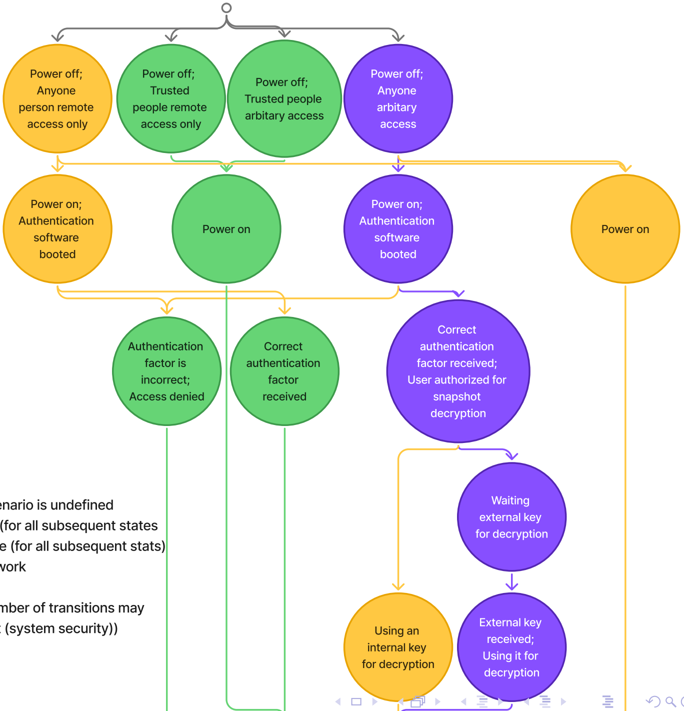
Use Cases & Requirements

Snapshot must remain **encrypted** at all times.

Requirements:

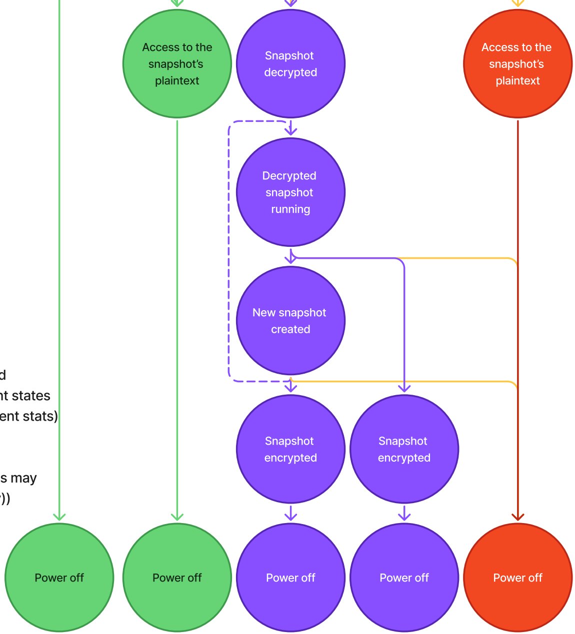
1. Mandatory authorization
2. Key stored on an **external USB drive**
3. Encryption of snapshots (AES, block-level)
4. Only encrypted data written to disk
5. Acceptable performance overhead ($\leq 2\times$)

Requirements 1 & 2 combined: the USB drive contains the key and serves as the authentication factor.



Legend:

- System state (computer - user)
- Note to the state
- In this state, the security of the scenario is undefined
- In this state the scenario is secure (for all subsequent states
- In this state the scenario is insecure (for all subsequent stats)
- This scenario is considered in our work
- Transition between states
- Transition between states (The number of transitions may vary and does not affect the final result (system security))



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Implementation: se_vault

There is a **file_vault** in Genode based on **Tresor** library:

- Tresor: AES encryption on 4K blocks, SHA-256 integrity, CoW atomicity
- file_vault: state orchestrator providing a secure FS session

Adapted file_vault → **se_vault**:

- GUI removed — headless operation
- Passphrase replaced by **USB key file**
- rump_fs/ext2 replaced by **lwext4/ext4**
- Added **fsck** crash recovery support with fsck
- Direct block session (skip image file creation)
- All block sizes standardized to 4K
- Graceful shutdown sequence added

Related Fixes & Adaptations

lwext4 bug fixes:

- `ftruncate`: pointer cast for correct file size
- `sync` call now propagated to lower VFS layers
  (critical — Tresor updates its state hash on `sync`)

Shutdown sequence (isomem / snapper / VFS):

- isomem: power-off button triggers final snapshot
- snapper, VFS: client counters trigger graceful shutdown
- lwext4 unmounted cleanly; mount flag cleared for fsck

Optimization: skipping image file pre-allocation
reduced initialization time from ≈20 min to ≈1 min.

Testing

Environment: QEMU with `-cpu host` (enables AES/SHA hardware acceleration).
2 GB image for isomem, 5 GB image for se_vault (4 GB user FS).

Functional tests (all passed):

- Normal boot / shutdown / restore cycle
- Encrypted image cannot be mounted without key
- Key/hash replacement attack — superblock not detected
- Crash recovery via fsck

Performance: first snapshot creation and restore, 5 runs each.

Results

Operation	Without enc.	With enc.	Overhead
Snapshot creation	2.10 s	51.00 s	$\approx 24\times$
Snapshot restore	2.50 s	27.30 s	$\approx 11\times$

Root cause: **known Tresor library limitation** — not optimized for multithreading. Acknowledged by the author; unresolved.

Evaluation

4 of 5 requirements met.

- ✓ Mandatory authorization (USB key)
- ✓ Key on external medium
- ✓ AES block-level encryption
- ✓ Only encrypted data on disk
- $\times$ Overhead $\leq 2\times$ *(Tresor limitation)*

Security gain: snapshot cannot be mounted or read without the key.
Integrity: SHA-256 at block level (Tresor) + hash at FS level (Snapper).

Conclusion

Adapted **file_vault** $\rightarrow$ **se_vault** for encrypted snapshot storage in PhantomOS on Genode.

First security research specifically for PhantomOS.
No analogous component existed prior to se_vault.

Future work:

- Fix Tresor performance (or switch to FS-level encryption via OpenSSL)
- Encrypt the isomem block device using se_vault
- Investigate merging Snapper and Tresor snapshot concepts